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Modelling of the Twin Rotor MIMO System (TRMS) Using the First Principles Approach

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ABSTRACT—The didactic ubiquitous rotorcraft, the Twin-Rotor Multi-Input Multi-Output System (TRMS) model, a universal laboratory apparatus/hardware is employed in this aerodynamics/ aeronautics modelling research work. From the survey conducted revealed that several advanced modelling techniques/approaches have been employed to control its flight/motion and operation. But due to a significant cross-linkage dynamics existing between the rotating twin rotors, cause the TRMS to be highly nonlinear, high-dimensional and complex, thus posing various control and modelling challenges. The complexity of this problem is comparable and analogous to a real-life helicopter. An optimal model-based solution to this has been presented here, based on the Newtonian and Lagrangian laws for nonlinear optimal control.

Key Words: Aerodynamics, Mathematical Modelling, MIMO, Optimal Control, TRMS

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1.0 INTRODUCTION

The Twin Rotor Multiple-Input Multiple Output (MIMO) System abbreviated as TRMS is an electromechanical laboratory equipment that comes in handy as a paradigm in the study of flying vessels like the helicopter, aeroplane, spacecraft, Unmanned Autonomous Vehicles (UAV), etc. Although the TRMS does not fly into air in space, yet it resembles a true helicopter in many respects [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]. Though the TRMS is a 3-dimensional (3D) equipment, it operates in a maximum 2 Degrees of Freedom (2-DOF) motion/rotation, of pitch and yaw axes, corresponding to the vertical and horizontal planes or subsections, though it can be arbitrarily confined to operate in only One Degree of Freedom (1-DOF) of either the pitch or yaw plane (the beam's vertical plane rotation, that is about the horizontal x-axis, or the beam's horizontal plane rotation, that is about the vertical z-axis). It can also be completely prevented from moving or rotating in any axes by locking the 2-axes to have a locked-down scenario. An important helicopter study is its hovering mode, used in rescues, aircraft delivery, which is carried out using the 2-DOF flexibility of the TRMS [11].

For controlling the rotations and motions of the TRMS in any or all axes and planes, the traditional Proportional-Integral-Derivative (PID) controllers have been employed. It is less expensive to design and easy to maintain. PID tuning for control purposes is done via the Ziegler-Nichols tuning laws, suited for linear processes [12]. But with complex, high-dimensional (high-order), nonlinear delayed plants like the TRMS (up to 24th order [13]) with no a priori knowledge of their physical and internal dynamics, PID control performs very poorly [14, 2]. To utilize PID controllers, linearization must be applied to the nonlinear systems, around a rest position [12, 15]. Even if this is done there are no guarantees that the system will be controlled well enough, because a non-zero transfer function which is marginally stable is being added to the linearized model [12].

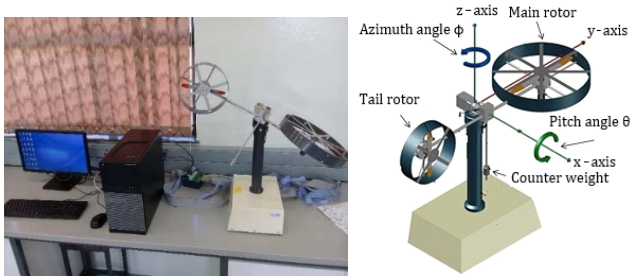
For optimal control of the rotations and motions of the TRMS in any or all axes/planes, the use of optimization schemes (approaches) and algorithms may be employed. In doing this, a modification of the PID controllers is done using numerous approaches. It spans from classical to intelligent control as follows:

1. Classical control (Root Locus techniques, Pole-Placement [16],Gain Margin and Phase Margin [16], Parametric Modelling [17], Model Reference [18], Model Predictive Control (MPC) [19], μ-synthesis [13], H∞ norm [20], Fractional Order Control [21], Integral

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Order Control [21], Trust Region Optimization [22], System Identification [23]),

2. Modern control linearization techniques with Linear Quadratic Regulator (LQR) [24], Linear Quadratic Gaussian (LQG) [24], Linear Parameter Varying (LPV) [25], Backstepping Control [26], Radial Basis Function [23], Dynamic Inversion [2], Wiener-Hammerstein, Decoupling control [27], Deadbeat Control [28], Sliding Mode Control (SMC) [29], Kalman Filtering [24], Embedded VHDL and FPGA [30], etc. to
3. Computational Intelligence control schemes, employing algorithms such as Fuzzy Logic Control (FLC) [31], FLC with LQR [32], Artificial Neural Networks (ANN) [33], Adaptive Neuro-Fuzzy Inference System (ANFIS) [34], Genetic Algorithms (GA) [35], Real-coded GA (RGA) [36], Genetic Programming (GP), Differential Evolution (DE) [37], Gravitational Search Algorithm (GSA) [38], Particle Swarm Optimization (PSO) [32], Artificial Bee Colony (ABC) [38], Ant Colony Optimization [39], Biological Cell Membrane control [40], Bacterial Foraging Optimization (BFO) [24], etc.



(a) (b)

Figure 1 (a) System Setup of the Twin Rotor MIMO System in the Measurements, Control and Instrumentation Laboratory of the Botswana International University of Science and Technology (BIUST), showing the interfacing and network cables (bluish) (b) TRMS Schematic and Notation [41, 2].

1.1 The TRMS Operation

The TRMS (shown in figure1 above) is a uniform beam clamped to a stand at its midpoint and contains twin shielded rotors (Main and Tail, seen in figure 1) fitted with propellers positioned at the ends of the beam. It is clamped to a stand containing a base wired for communication and data transfer. At rest, the beam is slanted at 60° to the horizontal due to the asymmetrical weights of the shielded propeller-fitted rotors (the Main rotor being heavier and larger in geometric size than the tail rotor, thus slanting the beam 60°). But when switched on, the motors on which each of the rotors with propellers (or blades) are attached begin to rotate orthogonal to each other. The rotations, cause the beam of the TRMS to be elevated or lowered in accordance with their angular velocities obtained through differentiation and filtration of the measured angles (positions) by the Real-time Kernel (TRK). Communication between the TRMS and the PC is through the Input/Output (I/O) cables, Network Interface Card (NIC) as well as RTK which works in MATLAB environment with WINDOWS operating system.

2.0 Mathematical Modelling of the TRMS

The dynamics of the TRMS is described to a high degree of accuracy, by the developed mathematical model (equations). These equations are wholly based on the motion laws of Newton and the Euler-Lagrange/Lagrarian equations.

Assumptions:

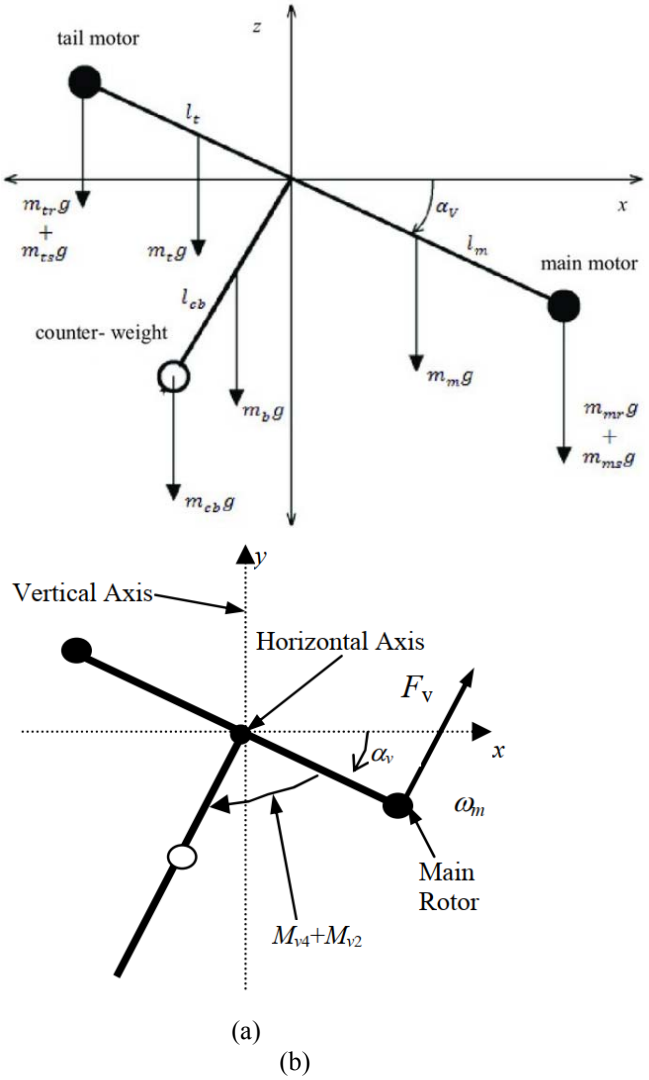
i. The frictional force is viscous friction
ii. The postulate of the theory of flow was used to describe the propeller air subsystem

NOMENCLATURE

M_{v1} – Restoring torque due to the gravitational forces
m_{mr} – Combined mass of Main rotor and its DC motor
m_{tr} – Combined mass of Tail motor and its DC motor
m_m – Beam's mass of Main portion
m_t – Beam's mass of Tail portion
m_{cb} – Counter-balance-weight mass
m_{ms} – Main shield mass
m_{ts} – Tail shield mass
l_m – Beam's Main portion length
l_t – Beam's Tail portion length
l_b – Length of the counterweight beam
l_{cb}– Distance from counter-balance-weight to the pivot joint
g – Acceleration due to gravity.
ω_m – Main rotor's angular (rotational) speed
ω_t – Tail rotor's angular (rotational) speed
$F_v(\omega_m)$ – Propulsive force dependency on the Main rotor's angular velocity
$F_h(\omega_t)$ – Propulsive force dependency on the Tail rotor's angular velocity
Ω_v – Beam's angular speed about the pitch-axis
K_v, k_h – Constants.
r_{ms} – Main shield's radius
r_{ts} – Tail shield's radius
M_h – Moments of forces about the z-axis
J_h – Moments of inertia relative to the z-axis
J_v – Summation of moments of inertia around the pitch axis

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J_{tr} – Moment of inertia of Tail rotor subsystem (Tail rotor + its DC-motor)
J_{mr} – Moment of inertia of Main rotor subsystem (Main rotor + its DC-motor)
S_v – Angular momentum in the vertical plane
S_h – Angular momentum in the horizontal plane
T_{mr} – Time constant of Main rotor and its DC motor subsystem
T_{tr} – Time constant of Tail rotor and its DC motor subsystem
α_v, α_h – The pitch angle, and tail angle of the beam.



(a)
(b)

Figure 2 Free-body diagram of the TRMS (a) Gravity forces and (b) Moments of Propulsive forces F_v and friction in the TRMS [20, 25, 30, 42, 43, 44, 45]

2.1 Determination of the moments of forces about the pitch (x)-axis

Using Newton's equations of motion (2nd law of motion) [42, 43, 44], summing moments of forces M_v about the horizontal or pitch axis (figure 2 above) in the TRMS beam yields:

$$M_v = J_v \frac{d^2\alpha_v}{dt^2} = \sum_{i=1}^{4} M_{vi} \quad (1)$$

$$J_v = \sum_{i=1}^{8} J_{vi} = m_{mr}l_m^2 + m_m\frac{l_m^2}{3} + m_{cb}l_{cb}^2 + m_b\frac{l_b^2}{3} + m_{tr}l_t^2 + m_t\frac{l_t^2}{3} + \left(\frac{m_{ms}}{2}r_{ms}^2 + m_{ms}l_m^2\right) + m_{ts}r_{ts}^2 + m_{ts}l_t^2 \quad (2)$$

The moment of forces due to gravity acting around the pitch axis of the beam M_{v1} is:

$$M_{v1} = g\left[\left(\frac{m_t}{2} + m_{tr} + m_{ts}\right)l_t - \left(\frac{m_m}{2} + m_{mr} + m_{ms}\right)l_m\right]\cos\alpha_v - \left(\frac{m_b}{2}l_b + m_{cb}l_b\right)\sin\alpha_v \quad (3)$$

2.2 Development of the propulsive forces on the beam.

The moment developed by the Main rotor's propulsive force M_{v2}, is given by:

$$M_{v2} = l_m F_v(\omega_m) \quad (4)$$

$$M_{v3} = -\Omega_h^2(A + B + C)\sin\alpha_v\cos\alpha_v \quad (5)$$

where the constants A, B and C are given as,

$$A = \left(\frac{m_t}{2} + m_{tr} + m_{ts}\right)l_t;\ B = \left(\frac{m_m}{2} + m_{mr} + m_{ms}\right)l_m;$$
$$C = \frac{m_b}{2}l_b + m_{cb}l_b$$

2.3 Azimuth Angle

The angular speed (velocity) about the yaw (z)-axis Ω_h, is:

$$\Omega_h = \frac{d\alpha_h}{dt} \quad (6)$$

and,
Moment due to friction M_{v4} is given as:

$$M_{v4} = -\Omega_v K_v \quad (7)$$

$$\Omega_v = \frac{d\alpha_v}{dt} \quad (8)$$

2.4 Determination of the moments of forces about the yaw (z)-axis

Similar to the beam's motion about the x-axis which is the vertical motion of the beam producing a pitch angle that generates moments of inertia and torques by the main rotor, the horizontal motion of the beam about the z-axis generate the following rotational moments and inertial moments:

$$M_h = J_h\frac{d^2\alpha_h}{dt^2} = \sum_{i=1}^{2} M_{hi} \quad (9)$$

2.5 Sum of components of inertial moments about the z-axis, J_h (dependent on pitch angle, α_v)

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$$J_h = \sum_{i=1}^{8} J_{hi} = \frac{m_b}{3} l_b^2 + m_{cb} l_{cb}^2 \cos^2 \alpha_v + \left[\left(\frac{m_m}{3} + m_{mr} + m_{ms}\right) l_m^2 + \left(\frac{m_t}{3} + m_{tr} + m_{ts}\right) l_t^2\right] \sin^2 \alpha_v + m_{ms} r_{ms}^2 + \frac{m_{ts}}{2} r_{ts}^2 \quad (10)$$

Consider figure 2(b) above:

$$M_{h1} = l_t F_h(\omega_t) \cos \alpha_v \quad (11)$$

Friction moment about the z-axis M_{h2} (which depends on the angular speed Ω_h) is:

$$M_{h2} = -\Omega_h \mathrm{k}_h \quad (12)$$

Combining together (1) – (8) and (9) – (12), dynamic equations of motion representing the helicopter motion of the TRMS in the vertical plane $\frac{dS_v}{dt}$ and the horizontal plane $\frac{dS_h}{dt}$ can be written as:

$$\frac{dS_v}{dt} = \frac{M_v}{J_v} = \frac{l_m F_v(\omega_m) - \Omega_v \mathrm{K}_v + g((A-B)\cos\alpha_v - C\sin\alpha_v) - \frac{1}{2}\Omega_h^2 (A+B+C)\sin 2\alpha_v}{J_v} \quad (13)$$

$$\Omega_v = \frac{d\alpha_v}{dt} = S_v + \frac{J_{tr}\omega_t}{J_v} \quad (14)$$

$$\frac{dS_h}{dt} = \frac{M_h}{J_h} = \frac{l_t F_h(\omega_t)\cos\alpha_v - \Omega_h \mathrm{K}_h}{J_h}$$

$$= \frac{l_t F_h(\omega_t)\cos\alpha_v - \Omega_h \mathrm{K}_h}{D\sin^2\alpha_v + E\cos^2\alpha_v + F} \quad (15)$$

$$\Omega_h = \frac{d\alpha_h}{dt} = S_h + \frac{J_{mr}\omega_m \cos\alpha_v}{J_h}$$

$$= S_h + \frac{J_{mr}\omega_m \cos\alpha_v}{D\sin^2\alpha_v + E\cos^2\alpha_v + F} \quad (16)$$

where D, E, F are constants, and are given by:

$$D = \frac{m_b}{3} l_b^2 + m_{cb} l_{cb}^2, \qquad E = \left(\frac{m_m}{3} + m_{mr} + m_{ms}\right) l_m^2 + \left(\frac{m_t}{3} + m_{tr} + m_{ts}\right) l_t^2, \qquad F = m_{ms} r_{ms}^2 + \frac{m_{ts}}{2} r_{ts}^2$$

[20, 25, 30, 42, 43, 44, 45]

The angular or rotational speeds are non-linear functions of the DC motors' input voltages u_{hh} and u_{vv}. Hence the following four additional equations:

$$\frac{du_{vv}}{dt} = \frac{1}{T_{mr}}(-u_{vv} + u_v) \quad (17)$$

$$\omega_m = P_v(u_{vv}) \quad (18)$$

$$\frac{du_{hh}}{dt} = \frac{1}{T_{tr}}(-u_{hh} + u_h) \quad (19)$$

$$\omega_t = P_h(u_{hh}) \quad (20)$$

Hence we should finally be having a set of 6 nonlinear equations for our mathematical model.

For the design simulations and model identification, the parameters $F_v(\omega_m), F_h(\omega_t), P_v(u_{vv}),\ P_h(u_{hh})$, which are all some nonlinear functions, must be determined. To do this, the phenomenological model was used. A polynomial approximation given in [45] based on some deep analysis in some other study is used in this research work as follows:

$$P_v(u_{vv}) = 90.99u_{vv}^6 + 599.73u_{vv}^5 - 129.26u_{vv}^4 - 1238.64u_{vv}^3 + 63.45u_{vv}^2 + 1283.41u_{vv} \quad (21)$$

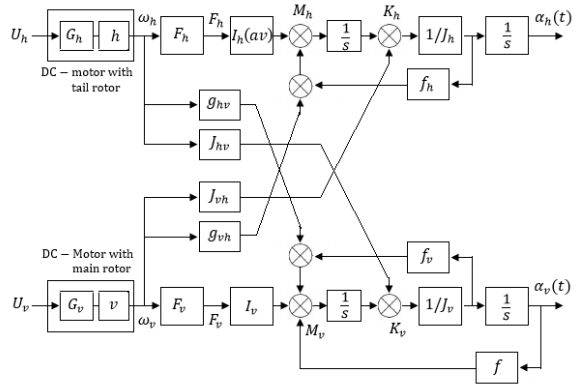
$$F_v(\omega_m) = -3.187 \times 10^{-12}\omega_m^5 - 4.096 \times 10^{-9}\omega_m^4 + 1.385 \times 10^{-6}\omega_m^3 + 1.234 \times 10^{-3}\omega_m^2 + 0.799\omega_m \quad (22)$$

$$P_h(u_{hh}) = 2020u_{hh}^5 - 194.69u_{hh}^4 - 4283.15u_{hh}^3 + 262.27u_{hh}^2 + 3796.83u_{hh} \quad (23)$$

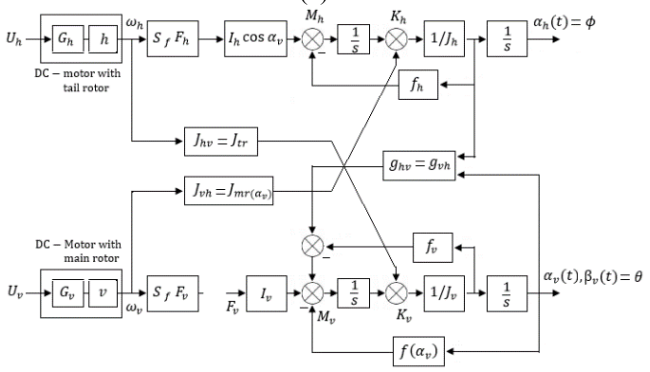
$$F_h(\omega_t) = 9.496 \times 10^{-13}\omega_t^5 - 9.844 \times 10^{-10}\omega_t^4 + 2.785 \times 10^{-7}\omega_t^3 - 1.73 \times 10^{-4}\omega_t^2 + 0.729\omega_t \quad (24)$$

2.6 Block Diagram Representations

The SIMULINK models in Matlab are used to represent the overall TRMS system in block diagrams. This has been presented in figures 3 as shown below:



(a)

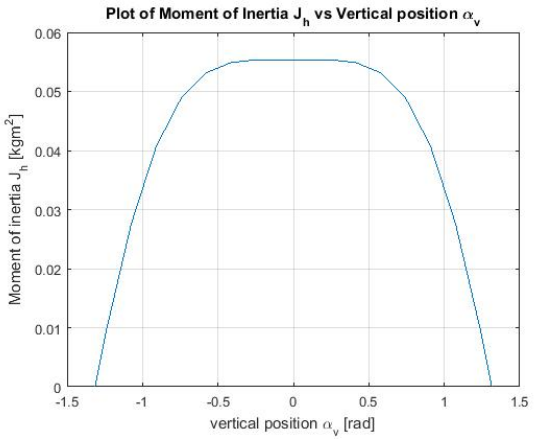


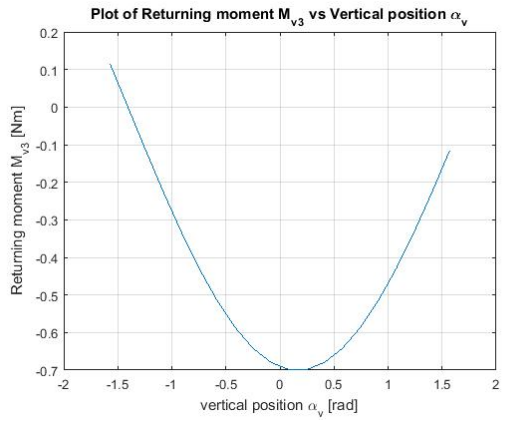
(b)

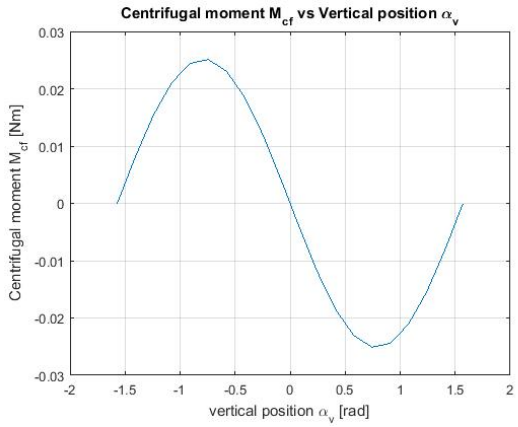
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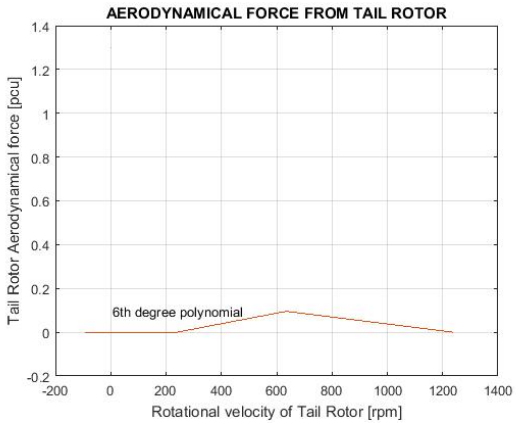
Figure 3 (a) MIMO Block diagram of the 1-DOF system (b) Complete MIMO block diagram of the 2-DOF TRMS **[3, 9]**

Simulations Results









DISCUSSION

From the matlab simulations presented, the angle of elevation (i.e. the pitch angle) is mainly influenced by the main rotor's rotation. The effect of the tail rotor's rotation is of lesser effect on the pitch, albeit modelled as a linear function to it. For the Azimuth or yaw angle, due to cross-couplings, the motion of the main rotor is highly difficult to incorporate in the model. However, attempts made with the moment M_{v2} as an exponential/growth function successfully handled the difficulty.

CONCLUSION

The 1-DOF and 2-DOF models are obtained from the phenomenological model of the TRMS presented in several papers and laboratory manual. The 1 & 2-DOF of the TRMS reveal the dynamics in 1-D and 2-D respectively. The underlying design of all model-based designs of the TRMS are based on the equations of the phenomenological model presented in (1)-(16)

The TRMS is an indispensable control system for control and optimization purposes. It represents the current modern control challenges in the aviation, aeronautic and aerospace industry and so widely used (popularized) and ubiquitous all over the world. It is employed as a standard benchmark to explore, invent, investigate, test, validate, standardize and implement in real time flight control schemes, approaches and methodologies. It is a highly nonlinear, high-dimensional multivariable control system that is difficult to manage and

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control. This is necessitates it being classed among advanced control systems and designs owing to these complexities.

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